

HANSCOM AFB, MA, USA

WMO: 725059

Lat: 42.470N

Lon: 71.289W

Elev: 133

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14702

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.3	(c) 7.8	(d) -11.4	(e) 3.0	(f) 7.5	(g) -7.4	(h) 3.7	(i) 11.4	(j) 25.4	(k) 24.7	(l) 23.4	(m) 26.2	(n) 6.2	(o) 290	(p) 0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.9	(c) 90.8	(d) 73.2	(e) 88.0	(f) 71.8	(g) 84.8	(h) 70.2	(i) 76.3	(j) 86.2	(k) 74.8	(l) 83.3	(m) 73.2	(n) 80.7	(o) 9.8	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	123.9	80.2	72.2	119.8	79.1	70.8	114.1	77.7	39.7	85.9	38.2	83.5	36.7	80.8	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	18.8	16.8	-4.2	95.5	6.5	3.2	-8.9	97.8	-12.7	99.7	-16.4	101.5	-21.1	103.8
			-5.1	79.1	6.2	1.7	-9.5	80.3	-13.1	81.3	-16.6	82.3	-21.1	83.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	26.6	28.4	36.2	47.7	57.6	66.4	72.5	71.0	63.4	51.6	41.8	32.1
	DBStd	17.96	10.76	9.38	9.32	8.57	7.78	7.04	5.68	5.82	7.33	8.08	8.56	9.25
	HDD50	2845	728	607	440	143	15	0	0	0	1	78	273	560
	HDD65	6232	1192	1025	893	524	257	66	7	14	115	422	696	1021
	CDD50	2742	1	2	12	74	250	493	699	650	404	127	26	4
	CDD65	655	0	0	0	5	27	109	241	199	68	6	0	0
	CDH74	6187	0	0	5	89	363	1053	2301	1771	556	49	0	0
	CDH80	2082	0	0	0	29	115	333	859	586	155	5	0	0
Wind	WSAvg	7.1	7.7	7.9	8.6	8.5	7.3	6.6	6.1	5.7	5.8	6.6	7.0	7.2
Precipitation	PrecAvg	46.0	3.9	3.5	4.1	3.9	3.7	3.5	3.5	3.5	3.8	4.0	4.4	4.4
	PrecMax	61.2	13.2	7.5	11.5	9.3	9.9	12.7	8.8	8.4	10.2	13.3	9.4	8.4
	PrecMin	27.7	0.7	0.2	0.6	0.8	0.9	0.2	0.5	0.8	0.9	0.6	0.7	1.0
	PrecStd	7.3	2.4	1.6	2.1	1.9	1.9	2.7	1.7	1.9	2.2	2.7	2.1	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.2	62.1	71.4	84.3	88.5	92.2	94.9	93.3	90.5	79.4	69.2	63.5
		MCWB	56.4	53.1	55.1	62.8	67.4	73.8	75.9	75.3	73.2	65.1	60.8	55.8
	2%	DB	53.1	52.2	62.6	74.3	83.1	87.8	91.0	89.8	84.1	74.0	64.4	55.8
		MCWB	47.9	45.4	51.0	57.4	65.8	72.3	73.5	73.4	69.9	64.5	57.5	51.4
	5%	DB	46.4	46.4	55.7	68.4	77.5	83.7	88.2	86.2	80.5	70.1	60.9	50.9
		MCWB	41.9	40.2	46.1	55.0	63.0	69.2	72.5	71.2	68.1	62.0	54.7	46.5
	10%	DB	41.3	42.0	50.3	63.2	72.6	80.6	84.7	82.8	76.3	65.7	56.6	46.0
		MCWB	36.5	36.2	42.8	51.9	60.7	67.4	70.9	69.9	66.2	58.6	50.9	41.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	55.8	58.5	65.5	71.3	76.7	79.1	78.2	75.4	70.2	64.2	58.7
		MCDB	59.1	61.0	66.8	76.0	84.9	88.0	91.1	88.8	83.5	74.5	67.3	61.4
	2%	WB	49.0	46.6	53.4	61.0	68.4	74.5	76.5	76.1	72.8	66.8	60.0	52.9
		MCDB	51.8	50.2	60.5	72.0	78.3	84.3	86.7	85.1	80.0	72.2	63.2	55.6
	5%	WB	41.9	41.0	47.6	57.6	65.7	72.3	74.8	74.4	71.1	63.3	55.8	47.2
		MCDB	46.5	45.5	54.3	65.8	74.2	80.1	83.7	81.9	76.9	68.7	59.3	50.3
	10%	WB	37.0	37.0	43.5	53.2	62.8	70.1	73.1	72.8	69.1	59.9	51.6	42.4
		MCDB	40.5	41.0	49.3	60.9	70.0	77.0	81.0	79.4	74.3	64.6	55.7	45.9
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.9	18.5	20.8	21.0	20.7	20.9	21.1	21.5	20.3	18.4	16.2
		MCDBR	21.2	23.1	26.7	29.3	28.8	25.1	23.9	23.9	24.3	24.7	23.0	21.6
		MCWBR	17.4	17.2	17.0	15.8	14.6	12.0	9.7	10.0	12.0	15.3	17.0	18.6
	5% WB	MCDBR	20.3	21.0	24.2	25.4	24.7	22.0	20.3	19.6	19.4	21.2	19.6	20.2
		MCWBR	17.9	17.3	17.6	16.4	13.9	12.1	9.7	9.9	11.7	15.4	16.8	19.3
Clear-Sky Solar Irradiance	taub	0.287	0.298	0.324	0.367	0.404	0.446	0.466	0.431	0.372	0.332	0.308	0.291	
	taud	2.485	2.472	2.472	2.385	2.321	2.233	2.185	2.265	2.411	2.537	2.565	2.560	
	Ebn at Noon	274	289	293	286	276	264	257	263	272	273	266	264	
	Edh at Noon	23	27	31	36	40	44	45	41	33	25	21	20	
All-Sky Solar Radiation	RadAvg	568	842	1156	1477	1631	1765	1868	1680	1352	890	613	467	
	RadStd	34	67	79	158	191	170	114	79	94	79	49	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	N/A
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+76

Nomenclature: See separate page